

Business Analytics for Audit and Fraud Detection: An Integrative Review of Digital Tools, a Red-Flag Taxonomy and a Conceptual AI-Assisted Prioritisation Framework

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Abstract—Business analytics (BA) and AI are increasingly proposed to strengthen audit and fraud detection, yet evidence on digital audit tools, red-flag indicators and the organisational conditions under which analytics delivers value remains fragmented across technical, accounting and practitioner literatures. This paper reports an integrative review that synthesises this fragmented evidence and develops a coherent conceptual framework linking audit-analytics capabilities, a red-flag indicator taxonomy and an AI-assisted prioritisation procedure. Using an integrative-review design with a PRISMA-informed screening protocol, 84 sources—peer-reviewed articles, industry reports and professional publications published between 2015 and 2025—were coded against five audit-model dimensions and synthesised thematically. The synthesis indicates that analytics can extend audit coverage from sampling to population-level testing and can support continuous and predictive assurance, while reported benefits depend strongly on data quality, analytical skills and governance, which the reviewed literature identifies as recurring obstacles. The review offers three synthesised contributions: (i) a five-cluster red-flag taxonomy aligned with corresponding analytical techniques; (ii) a comparative synthesis of five audit-analytics platforms and an analytics-maturity model; and (iii) a model-agnostic, conceptual AI-assisted red-flag prioritisation framework expressed as a weighted risk score and a reviewable prioritisation procedure. The prioritisation framework is conceptual: it is a synthesis-driven proposal and is not empirically validated in this study, and no accuracy, precision or recall figures are claimed for it. For practitioners, the synthesis provides a structured basis for tool selection, for sequencing analytics maturity and for governing algorithmic outputs; for researchers, it identifies empirical validation of such frameworks as an open priority.

Index Terms—Business Analytics, Audit Automation, Fraud Detection, Red-Flag Indicators, Integrative Review, AI-Assisted Prioritisation, Analytics Maturity

I. Introduction

The rapid digitalisation of financial systems has reshaped auditing by generating volumes of accounting data that traditional audit methods cannot readily process [1]. Conventional auditing, based on periodic review cycles, human judgment and selective sampling, was designed for a less complex era in which financial records were fewer, more stable and less prone to manipulation [2]. In modern organisations, manual audit methods typically

scrutinise only a limited subset of transactions, which allows misstatement in large transaction populations to remain undetected for prolonged periods and, in some cases, to be discovered only after significant loss [3]. These methods also struggle to detect subtle, deliberately concealed alterations in financial records and therefore may not provide the timely assurance that contemporary business cycles require. As financial crime becomes harder to identify because of digital payment systems, complex vendor networks and dispersed data, conventional audit methods may become a governance risk in themselves [4], [5].

Business analytics (BA) has been proposed as a response to these structural constraints, shifting assurance from reactive and periodic testing towards proactive, continuous and data-driven practice [6], [7]. By enabling examination of complete transaction populations rather than samples, BA allows auditors to surface anomalies, behavioural irregularities and structural inconsistencies at scale. Machine learning, predictive risk models, network analysis and natural language processing have been reported to extend auditors' detection capabilities to indicators such as irregular journal entries, atypical vendor relationships and after-hours access, which may precede or accompany fraudulent activity [8]–[10]. The reviewed literature reports that organisations adopting analytics-driven auditing tend to observe improved detection outcomes relative to conventional techniques, alongside reductions in false positives attributable to machine-learning-based anomaly detection [4], [11]; the magnitude of these gains varies considerably with data quality and deployment context.

Despite this promise, existing research on BA in auditing remains fragmented in several respects. There is limited comparative evidence on the relative effectiveness of different analytical methods—for example supervised versus unsupervised learning—across audit contexts and fraud types. Guidance on practical implementation is sparse, particularly regarding change management, workflow integration and the hybrid competencies that combine

accounting expertise with data-science skills. Moreover, much of the literature adopts narrow perspectives, examining a specific technology, a single industry case or a technical performance metric in isolation from the organisational and ethical challenges of implementation [12]. As a result, practitioners face uncertainty about tool selection, integration into existing audit processes, algorithmic bias and transparency, and the consequences of analytics for audit quality and professional scepticism. A synthesis that connects these strands is therefore needed.

These gaps carry implications for several stakeholder groups, which motivates this review. The absence of established implementation standards leads to inconsistent adoption and can undermine confidence in algorithmic outputs, while the lack of uniform standards for algorithmic auditing poses regulatory challenges as analytics spreads across global assurance practice. The gap between current curricula and the hybrid skills auditors require affects graduates' readiness for data-driven practice [13]. As expectations and oversight increase, analytical capability becomes important for maintaining the relevance and value of audit operations [14]. Finally, the scarcity of a connected conceptual framework means that empirical studies in this area proceed without a common structure, limiting comparability.

Against this background, this review addresses four research questions:

- RQ1: Which analytical methods and digital tools does the reviewed literature apply to audit and fraud detection, and with what reported capabilities and limitations?
- RQ2: How can red-flag indicators be classified into a coherent taxonomy that links each class to appropriate analytical techniques?
- RQ3: How do analytical maturity, tool choice, sector and organisational readiness condition the realisation of analytics benefits in audit?
- RQ4: What conceptual framework can integrate a red-flag taxonomy, analytics capabilities and AI-assisted prioritisation into the audit workflow, and what remains to be empirically validated?

To address these questions, the study undertakes a qualitative integrative review of 84 sources drawn from peer-reviewed articles, industry reports and professional publications published between 2015 and 2025. The synthesis suggests that although analytics offers demonstrable advantages in coverage and timeliness, its effective application is constrained by data-quality deficiencies repeatedly identified as the primary obstacle [15], skill gaps within audit teams [16] and ethical concerns surrounding AI-generated insight and imbalance [17]. Adoption differs across sectors: regulated financial institutions are among the most advanced adopters, driven largely by regulatory pressure, whereas small and medium-sized enterprises lag because of resource and capacity constraints [18],

[19]. Network analysis is reported to strengthen forensic investigation [20], and regulators increasingly use analytics for systemic risk monitoring and compliance trend analysis [21].

The review makes the following contributions:

- C1 (methodological): a documented, PRISMA-informed integrative-review protocol (search, screening, coding and triangulation) that makes the synthesis and its limitations auditable, with points requiring author confirmation explicitly flagged.
- C2 (conceptual): a five-cluster red-flag taxonomy in which each class is defined and linked to the analytical techniques that are reported to detect it, distinguishing categories conceptually so that they do not substantially overlap.
- C3 (comparative): a synthesis of five audit-analytics platforms across shared dimensions together with an analytics-maturity model that connects descriptive-to-prescriptive capability with data, technology, workforce, governance and explainability conditions.
- C4 (framework): a model-agnostic, conceptual AI-assisted red-flag prioritisation framework, expressed as a weighted risk score and a reviewable procedure, positioned explicitly as a proposal for future empirical testing rather than a validated model.

The remainder of the paper is organised as follows. Section II describes the review methodology. Section III synthesises the findings and discusses their implications. Section IV presents conclusions, policy implications and limitations, and identifies priorities for future research.

II. Methodology

This section describes the review protocol. It first sets out the theoretical lenses adopted, then the review design, search and screening strategy, coding and synthesis procedure, and finally the analytical frameworks used to organise the synthesis (an audit-model comparison, a tool categorisation, a red-flag taxonomy and a conceptual prioritisation framework).

A. Theoretical Foundations

Three established theoretical lenses guide the synthesis. First, the Fraud Triangle explains fraud as the convergence of pressure (incentive), opportunity and rationalisation [66]. Audit analytics bears primarily on the opportunity element, by continuously exposing control weaknesses and anomalous transactions, and secondarily on pressure, by supplying early-warning signals; the taxonomy in Section II-G is organised around observable indicators of these elements. Second, the Technology–Organization–Environment (TOE) framework is used to explain uneven adoption, distinguishing technological readiness, organisational capability (skills, governance, change management) and environmental factors such as regulation and competitive pressure [67]. Third, an analytics-maturity perspective, spanning descriptive, diagnostic, predictive

and prescriptive capability, provides a developmental scale against which reported detection capability is assessed in Section III-A. Together these lenses connect the reviewed tools and red-flag indicators to established explanations of fraudulent behaviour and of organisational technology adoption. They are used to structure interpretation; they are not tested empirically here.

B. Review Design and Analytical Strategy

The review follows an integrative design rather than a systematic review, because the aim is to synthesise heterogeneous evidence—empirical studies, conceptual frameworks, industry reports and practitioner guidance—into a coherent account, emphasising interpretation over statistical pooling. Integrative reviews are appropriate where the literature is methodologically diverse and where the objective is conceptual synthesis rather than effect estimation. The design is qualitative and interpretive: audit tools and red-flag indicators are treated as acquiring meaning and effectiveness through their position within particular institutional, technological and cultural configurations, rather than as context-free instruments. This interpretive stance is adopted deliberately and is reflected in the limitations in Section IV-C.

C. Search Strategy and Source Selection

A multi-stage retrieval strategy was used to assemble academic and professional literature on audit analytics and fraud detection. Thematic accuracy was pursued through purposive sampling with iterative refinement of search terms. Keyword groups spanned audit automation, fraud analytics, continuous auditing, anomaly detection in finance and digital auditing frameworks, together with related terms.

Sources were retrieved from five databases: Scopus ($n = 64$), Web of Science ($n = 41$), IEEE Xplore ($n = 36$), SSRN ($n = 58$) and Google Scholar ($n = 73$), a combined total of 272 records for the period 2015–2025. In addition, 25 industry reports and white papers issued by ISACA and the Big Four accounting firms were included to complement academic evidence with practitioner perspectives. Records were screened in two stages: title and abstract review, followed by full-text review. The arithmetic of the selection process is summarised in Table I and is internally consistent ($272 \text{ academic} + 25 \text{ professional} = 297 \text{ retrieved}$; $25 \text{ duplicates removed} = 272$; $118 \text{ excluded at title/abstract} = 154$; $70 \text{ excluded at full text} = 84 \text{ included}$).

Inclusion required that a source (i) address digital auditing, audit analytics or fraud analytics; (ii) display sufficient methodological or procedural transparency to permit appraisal; and (iii) offer practical relevance to organisational or audit decision-making. Sources were excluded if they were editorial or non-substantive, lacked clarity in execution, or addressed non-financial topics. To avoid presenting academic and professional evidence as

TABLE I
Review Protocol and Source Selection

Stage	Count	Note
Records retrieved (5 databases)	272	Scopus, Web of Science, IEEE Xplore, SSRN, Google Scholar
Additional professional sources	25	ISACA and Big Four reports
Total initially retrieved	297	
Duplicates removed	25	Overlap across databases
Records after de-duplication	272	Eligible for screening
Excluded at title/abstract	118	Out of scope or non-substantive
Excluded at full text	70	Editorial, low rigor, or off-topic
Included in synthesis	84	Tagged by source type

equivalent, sources were tagged by type at the coding stage so that claims resting on grey literature could be distinguished from those supported by peer-reviewed research.

D. Coding, Synthesis and Validation Protocol

Coding followed the three-stage thematic-synthesis approach of Thomas and Harden: line-by-line coding of each source to extract features, red-flag indicators, implementation challenges and value claims; consolidation of initial codes into descriptive themes; and development of analytical themes that connect these into an interpretive framework. Coding was iterative, and the codebook was reviewed and refined across analytical cycles to improve thematic consistency across document types. Further sources were added only until new themes ceased to emerge.

Findings were triangulated by contrasting academic evidence with the practices reported by the Big Four firms, by checking claimed tool functionality against publicly available documentation, and by relating red-flag typologies to case narratives in forensic accounting reports. Triangulation was used to strengthen interpretation; it does not convert the synthesis into an empirical validation.

E. Comparative Framework for Audit Models

Audit models were compared along five dimensions: data coverage, audit timing, detection responsiveness, resource intensity and fraud traceability. These dimensions were derived from the reviewed literature and industry guidance and served as coding categories. Table II contrasts conventional and analytics-enabled auditing along these dimensions. Each included source was assessed for alignment with this framework, which provided a consistent basis for comparison and for identifying transition patterns from conventional to analytics-enabled practice.

TABLE II
Comparative Framework: Conventional versus Analytics-Enabled Auditing

Dimension	Conventional Auditing	Analytics-Enabled Auditing
Data coverage	Sample-based, with restricted visibility.	Population-level testing of complete datasets [22].
Audit timing	Periodic, typically annual or quarterly.	Continuous or near-real-time procedures [23].
Detection response	Post-event and reactive.	Real-time or predictive identification for pre-emptive action [24].
Resource intensity	Labour-intensive and manual.	Automation can reduce manual effort [60].
Fraud traceability	Limited, owing to fragmented records.	Improved, through auditable digital records [25].

F. Tool Categorisation and Functional Analysis

Five platforms frequently cited in academic and commercial sources were examined: ACL, IDEA, Power BI, Tableau, and open-source analytics (Python and R, treated together as one category because the reviewed sources discuss them jointly). Table III compares them across dimensions shared by the sources: primary audit application, data integration, visualisation, fraud/anomaly detection support, ease of adoption and principal limitation. The comparison is deliberately restricted to dimensions on which the reviewed sources report; no claims are made about capabilities outside this evidence base.

The comparison suggests a trade-off between analytical power and accessibility: open-source platforms are reported as the most flexible for complex fraud analytics, whereas ACL, IDEA, Power BI and Tableau trade some analytical depth for usability, structured reporting or visual exploration. No single platform was reported as universally preferable; selection appears to depend on the audit objective, the team’s competencies and the available data infrastructure [12].

G. Red-Flag Indicator Taxonomy

Red-flag indicators are observable signals that warrant audit attention. This review organises them into five clusters that are intended to be conceptually distinct: transaction irregularities, behavioural anomalies, structural anomalies, documentation discrepancies and relational fraud patterns. Unlike a simple list, each cluster is defined, illustrated, linked to the analytical techniques reported to detect it, and characterised by its limitations (Table IV). Figure 1 presents the taxonomy conceptually. The clusters are not assumed to be exhaustive, and indicators may co-occur across clusters—an observation that motivates the combined prioritisation framework in Section II-I.

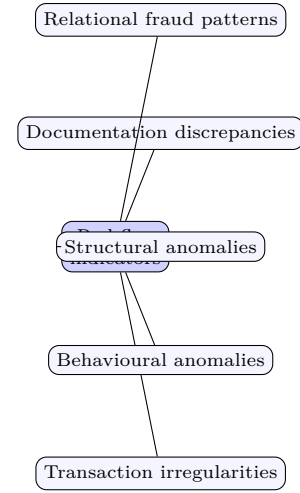


Fig. 1. Five-cluster red-flag indicator taxonomy (conceptual synthesis; clusters are non-exhaustive and may co-occur).

H. Analytics-Driven Audit Workflow

The reviewed sources converge on a five-stage analytics workflow for audit: data collection, data preparation, exploratory modelling, risk scoring and red-flag issuance [58]. Figure 2 presents this workflow conceptually. Data collection draws structured and unstructured financial data from enterprise resource planning systems, spreadsheets, transaction logs and internal communications. Data preparation—cleansing, de-duplication, transformation and normalisation—ensures that datasets are consistently formatted for analysis; the reviewed sources describe this stage as the most resource-intensive of the workflow. Exploratory modelling applies clustering, trend detection and outlier identification to surface latent relationships and irregularities [59]. Risk scoring assigns weighted values to the frequency and severity of anomalies to rank investigative tasks, and red-flag issuance produces alerts and visual summaries for auditor dashboards. The workflow is iterative rather than linear: insights from red-flag outputs feed back to refine data features and model parameters, and governance mechanisms align analytical activity with organisation-wide risk protocols.

I. AI-Assisted Red-Flag Prioritisation Framework

Building on the workflow model and the taxonomy, this review consolidates the reported techniques into a conceptual, model-agnostic prioritisation framework. The purpose is to give a single interpretable score to each transaction or journal entry so that finite audit resources can be directed towards the most consequential cases. Let x_r denote the feature vector of record r (temporal, monetary, categorical and textual attributes). Four terms are combined, each normalised to $[0, 1]$:

- $a(x_r)$: an unsupervised anomaly score (for example from an isolation forest, an autoencoder or a clustering-distance measure);

TABLE III
Comparative Analysis of Audit-Analytics Platforms (Open-Source = Python/R)

Platform	Primary audit application	Data integration	Visualisation	Fraud / anomaly detection	Ease of adoption	Principal limitation
ACL	Structured audit tests and standardised reporting	Moderate, diverse datasets	Basic	Moderate, for irregularities	Easy for basic use	Limited advanced detection [26]
IDEA	File-consistency checks, fuzzy matching, regression-based anomaly identification	High, across formats	Adequate	Strong suitability for fraud identification	Moderate, training needed	Less flexible analytic workflows [27]
Power BI	Interactive financial dashboards and visual exploration	Robust, multi-source	High, interactive	Moderate; no native fraud models	Relatively easy	Limited inherent fraud detection [28]
Tableau	Visual analytics of financial data	Robust, seamless	Very high	Limited direct utility for fraud detection	Intuitive	Limited inherent fraud detection [29]
Python / R	Custom, reproducible analytics and modelling	Strong and scalable	Customisable	High, for complex analytics (reported)	Steep learning curve	Requires programming skill; no built-in audit interface [30]

TABLE IV
Five-Cluster Red-Flag Indicator Taxonomy

Cluster	Definition (what it detects)	Representative indicators	Analytical techniques reported	Audit relevance	Limitations
Transaction irregularities	Deviations in timing, frequency, duplication or value of transactions	Duplicate payments; out-of-hours postings; unusual amounts	Rule-based tests; frequency and outlier detection; clustering	Detects possible control bypass in transaction processing	Prone to false positives; sensitive to threshold calibration
Behavioural anomalies	Deviations in user or entity behaviour relative to peers or history	Recurrent control violations; irregular access; resistance to oversight	Behavioural analytics; peer-group comparison; sequence mining	Early indication of insider or override risk	Requires reliable access-log data; privacy considerations
Structural anomalies	Inconsistencies in ledgers, reconciliations or reporting structures	Unbalanced accounts; incongruous reconciliations; unit inconsistencies	Statistical reconciliation; graph analysis of accounts	Signals weak control design and governance gaps	Requires integrated, well-modelled data
Documentation discrepancies	Manipulated, duplicated or unauthorised documents	Forged or duplicated invoices; unauthorised edits; altered dates	Duplicate and near-duplicate detection; OCR with NLP; metadata analysis	Addresses invoice authorisation and documentary integrity	Sensitive to data quality; can be concealed adversarially
Relational fraud patterns	Coordinated or collusive relationships among entities	Bid-rigging; related-party networks; circular payments	Graph/network analytics; community detection; link analysis	Surfaces schemes that evade rule-based tests	Data-intensive; interpretability and false-link risks

- $b(x_r)$: a behavioural-deviation score relative to a peer group or the entity's own history;
- $c(x_r) \in \{0,1\}$: a binary indicator that record r violates a defined control rule;
- $f(x_r)$: a normalised frequency/severity term capturing how often and how materially such an event occurs.

The composite risk score is

$$s(r) = w_1 a(x_r) + w_2 b(x_r) + w_3 c(x_r) + w_4 f(x_r), \quad (1)$$

where the weights satisfy $w_i \geq 0$ and $\sum_{i=1}^4 w_i = 1$. Because all terms are normalised to $[0, 1]$ and the weights sum to one, $s(r) \in [0, 1]$, which admits a single interpretable threshold τ . Records with $s(r) > \tau$ are escalated for investigation, while records below τ enter routine monitoring; in practice τ should be calibrated against the auditor's relative tolerance for false positives and false negatives rather than fixed a priori. The weights

encode the auditor's emphasis and risk appetite and, like τ , require calibration on representative data. Algorithm 1 summarises the procedure, which operationalises the anomaly-detection, risk-scoring and red-flag-issuance stages of the workflow. Figure 3 presents the framework conceptually.

The framework is intentionally model-agnostic: each term can be instantiated with statistical, clustering, graph-based or supervised techniques reported in the literature, allowing model complexity to be matched to an organisation's data and analytical maturity. Crucially, the framework is conceptual and has not been validated in this review; no accuracy, precision or recall figures are claimed for it, and its component techniques are reported individually in the reviewed literature rather than as an integrated, evaluated system. Empirical testing is identified as a future priority in Section IV-C.

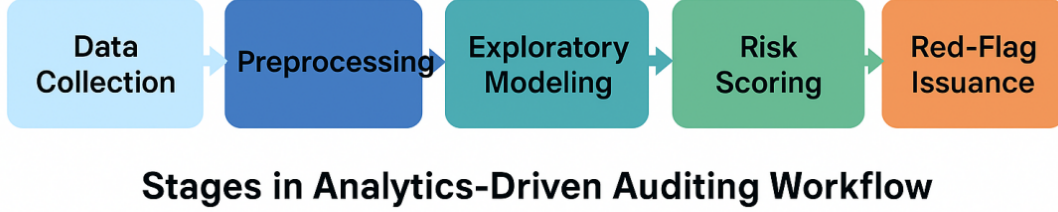


Fig. 2. Analytics-driven audit and fraud-detection workflow (five conceptual stages; described by the reviewed sources as the most resource-intensive phase, data preparation precedes modelling, scoring and red-flag issuance).

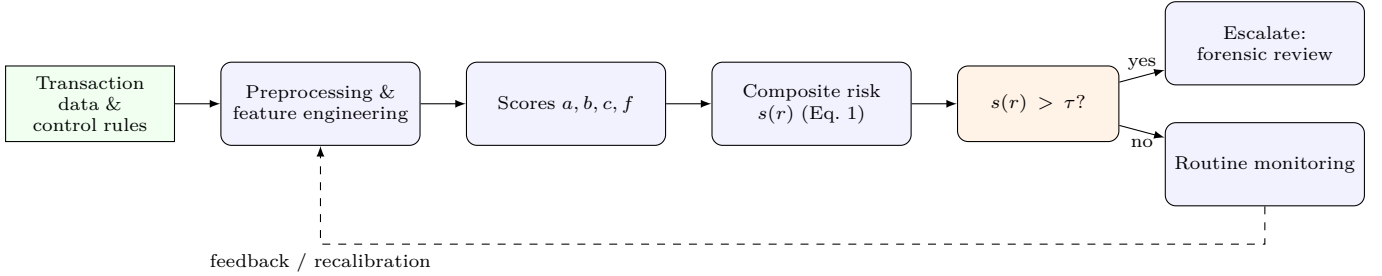


Fig. 3. Conceptual AI-assisted red-flag prioritisation workflow (proposed synthesis; not empirically validated). Records above τ are escalated, others monitored, and outcomes feed back to recalibrate scores and weights.

III. Results and Discussion

This section synthesises the reviewed evidence around RQ1–RQ4. Throughout, we distinguish between what the reviewed sources report and our own conceptual interpretation.

A. Analytical Maturity and Capability

Reported detection capability appears to be associated with analytical maturity. The reviewed literature distinguishes descriptive analytics (summarising historical patterns), diagnostic analytics (explaining the drivers of anomalies), predictive analytics (anticipating likely misconduct) and prescriptive or AI-assisted analytics (recommending actions, often using machine learning) [26], [31]. Table V summarises these levels and the conditions on which they depend. Maturity is not merely a

function of tooling: it is coupled with data availability and quality, workforce skills, governance and the explainability of models [32], [33]. Several sources caution that progress towards predictive and prescriptive analytics requires infrastructure investment, skilled staff and change-management effort, and that organisations in diagnostic or early predictive stages may struggle to justify prescriptive systems [34]. This suggests that maturity should be treated as a socio-technical rather than a purely technological construct—an interpretation consistent with the TOE lens adopted in Section II.

B. Tool-Specific Outcomes

The synthesis of platform evidence suggests that tool effectiveness is contingent rather than absolute. Open-source analytics (Python/R) is reported to offer the

Algorithm 1 AI-Assisted Red-Flag Prioritisation (Conceptual)

Require: transaction data D , control rules C , weights w , threshold τ

Ensure: ranked red-flag list L

preprocess D and engineer features (temporal, monetary, categorical, textual)

for each record $r \in D$ do

 compute anomaly score $a(x_r)$ and behavioural score $b(x_r)$

 set $c(x_r)$ from control-rule engine C

 compute composite score $s(r)$ via Eq. (1)

end for

rank D by $s(r)$ in descending order

for each record r with $s(r) > \tau$ do

 assign red-flag cluster from the taxonomy (Table IV)

 emit alert to auditor dashboard for review

end for

return L

TABLE V

Analytics Maturity Levels and Their Reported Contribution to Audit/Fraud Workflows

Level	Contribution to audit/fraud workflow	Principal enablers / constraints
Descriptive	Retrospective reporting and population-level summarisation	Data availability; basic tooling
Diagnostic	Root-cause analysis of control failures and anomalies	Integrated, well-modelled data; analytical skills
Predictive	Risk-based prioritisation and early warning	Curated data; model validation; governance
Prescriptive / AI-assisted	Continuous prioritisation and response recommendation	Explainability; skills; change management

greatest flexibility for complex fraud analytics and to be particularly applicable in high-risk contexts such as finance and e-commerce, where schemes may outpace rule-based detection [35], [36]. Visualisation-oriented platforms (Tableau, Power BI) are reported to support exploration and communication of financial patterns, but the reviewed sources note that they lack inherent fraud-detection capability [29]. IDEA occupies a pragmatic middle ground for file-consistency and regression-based anomaly identification, and ACL is reported as reliable for structured audit procedures because of its usability and standardised reporting, despite limited advanced detection [30]. Effectiveness is therefore conditioned by technical proficiency, data quality and the complexity of the threat [12], [37], and combining tools may complement their individual limitations [38]. This finding supports the framework’s model-agnostic design, in which no single technique is privileged.

C. Red-Flag Indicators in Practice

The five-cluster taxonomy (Table IV) is grounded in the reviewed sources. Transaction irregularities are reported as the most frequently addressed class, commonly identified through rule-based and frequency/outlier analysis [39]. Documentation discrepancies—forged, duplicated or unauthorised documents—are reported as widespread, reflecting vulnerabilities in invoice authorisation and documentary integrity [40]. Behavioural anomalies are associated with sudden changes in conduct, irregular access or resistance to oversight, and are treated as early indicators of concealed misconduct [41]. Structural anomalies arise from inconsistencies across statements, business units or reporting structures and often point to isolated controls and governance gaps [42]. Relational fraud patterns, such as bid-rigging or collusive networks, are reported as the least prevalent and most elusive, requiring graph-based analytics and cross-entity tracking [43]. Because these classes can intersect—and intersecting signals may indicate elevated risk—the synthesis supports combining multiple analytical techniques and machine-learning classifiers augmented with domain knowledge to improve sensitivity and specificity [44], [45]. We note, however, that the association between specific indicators and realised fraud is reported rather than demonstrated causally in the reviewed literature.

D. Sectoral and Organisational Patterns

Adoption and effectiveness vary across sectors, influenced by regulation, risk exposure, technological capability and organisational priorities [47]. In highly regulated sectors such as finance, healthcare and telecommunications, adoption is generally more advanced, driven by compliance mandates that favour real-time monitoring and integrated platforms [48]. In lower-resource settings—manufacturing, retail and small to medium-sized enterprises—limited budgets, weak IT infrastructure and poorly integrated data restrict adoption, leaving such organisations reliant on basic visualisation or easily implemented tools that lack the analytical depth needed for timely detection [49], [50]. Public-sector and non-profit organisations face a distinct trajectory, remaining exposed to procurement irregularities and misappropriation while implementing analytics under funding and capacity constraints [51], [52]. Across these contexts, institutional readiness, data-governance culture and alignment with objectives appear to shape assimilation more than the mere availability of technology [53].

Comparative capability gaps between conventional and analytics-driven auditing are reported along several parameters: adaptability, staff proficiency, data infrastructure, detection speed and scalability [54]–[57]. Whereas conventional audits rely on accounting proficiency, analytics-driven audits require combined finance and data-science expertise; organisations without targeted recruitment or training may acquire tools but struggle

to extract diagnostic insight. Legacy, isolated systems compromise data integrity and limit anomaly detection, whereas integrated platforms with real-time data streams can support continuous auditing. Manual audits are bounded by human capacity, whereas analytics-driven systems offer modularity and scalability—although we interpret these advantages as reported tendencies rather than guaranteed outcomes.

E. Workflow Efficacy and Governance Alignment

Integrating analytics into audit phases is reported to accelerate procedural efficiency and responsiveness. The workflow comprises five interconnected phases—data collection, data preparation, modelling, risk assessment and red-flag generation—which function as a feedback loop rather than a linear sequence [58]. Data preparation is consistently reported as the most resource-intensive phase in terms of time and effort, and each of its steps influences model accuracy and performance [59]. Automation of routine preparation tasks, such as extract-transform-load pipelines and data-wrangling, may reduce manual effort and improve consistency [60]. Risk scoring is reported to prioritise audit activity by weightings of severity, frequency and potential financial impact, with cases above defined thresholds escalated for forensic investigation [61]. Effective governance alignment translates analytical outputs into institutional action: recurring indicators of procurement irregularities may prompt revisions to vendor-selection or approval processes [62], [63]. We interpret this as evidence that the value of analytics depends as much on governance and feedback mechanisms as on the analytical models themselves.

F. Synthesis of Implementation Challenges

The synthesis points to recurring implementation challenges that cut across sectors and tools (Table VI). Data quality and interoperability are repeatedly identified as the primary obstacle [15], with data-conversion failures capable of undermining detection accuracy [25]; skill gaps affect audit teams [16]; legacy infrastructure constrains integration [49]; and algorithmic bias, fairness and transparency raise governance concerns [13], [17], [46]. Because these challenges are organisational as well as technical, mitigation is likely to require data governance, hybrid competency development and explainable-AI practices in combination, rather than tooling alone. This interpretation is consistent with the sources reviewed, although none of the reviewed studies evaluates these mitigations jointly.

G. Validation and Forward Pathways

Triangulation across academic, professional and forensic sources supports the interpretation that analytical tools can improve audit outcomes, though it does not establish causal effects. The synthesis identifies three enduring constraints: data interoperability across platforms [64], the need for training ecosystems covering SQL, Python

and AI interpretability [65], and regulatory clarity on algorithmic auditing and explainable AI in assurance [13]. Cross-disciplinary collaboration—between audit, IT, data science, legal and compliance functions—is indicated as important, and controlled simulation environments (“fraud sandboxes”) are proposed as a means of testing algorithmic models without legal or ethical exposure. We present these as reported proposals and our own interpretation, not as validated interventions.

IV. Conclusion and Policy Recommendations

A. Conclusion

This review set out to synthesise fragmented evidence on business analytics for audit and fraud detection and to connect it into a coherent framework. Four things are established. First, the reviewed literature consistently associates analytics with expanded coverage (population-level testing), greater timeliness (continuous and predictive assurance) and improved traceability, relative to conventional sampling-based auditing—benefits that are reported rather than demonstrated causally in an integrative review. Second, the proposed contributions are a documented review protocol, a five-cluster red-flag taxonomy linked to analytical techniques, a comparative synthesis of five platforms with an analytics-maturity model, and a conceptual AI-assisted prioritisation framework. Third, for practitioners the synthesis offers a basis for sequencing analytics maturity, selecting tools according to objective and competency, and governing algorithmic outputs through explicit thresholds and feedback. Fourth, what remains unvalidated is the prioritisation framework itself: it is a synthesis-driven proposal whose components are reported individually in the literature but have not been evaluated as an integrated system, and no accuracy metrics are claimed for it. Accordingly, the central empirical task for future work is to test whether such a framework improves detection or triage in practice.

B. Policy Recommendations

Several implications follow for policymakers, regulators and audit leaders. First, standardising algorithmic-audit protocols would give regulators and practitioners a common basis for assurance; internationally accepted standards for AI and machine learning in auditing, with emphasis on transparent processes, are indicated [13]. Second, investment in digital audit infrastructure—upgrading legacy systems, adopting cloud-based platforms and centralised data governance—is a precondition for the integration of analytical tools. Third, competency development should be reinforced through a hybrid auditor-competency framework covering SQL, Python, inferential statistics and data visualisation, supported by professional certification and university programmes. Fourth, controlled fraud-simulation environments (“fraud sandboxes”) are proposed as regulated settings in which detection algorithms

TABLE VI
Recurring Implementation Challenges and Reported Mitigations

Challenge	Evidence reported in reviewed sources	Mitigation indicated in sources	Refs
Data quality and interoperability	Data-quality deficiencies repeatedly identified as the primary obstacle; conversion failures reduce accuracy	Standardised data governance; automated ETL/data-wrangling pipelines	[15], [25], [60]
Skill gaps in audit teams	Reported shortage of combined finance and data-science competencies	Hybrid competency frameworks; targeted training	[16], [55]
Legacy, fragmented infrastructure	Isolated, antiquated systems compromise data integrity and limit detection	Phased migration to integrated platforms; real-time data streams	[49], [53]
Algorithmic bias, fairness and transparency	Concerns about fairness and bias in high-stakes models; need for explainability	Bias testing; explainable AI; documented audit trails	[13], [17]
Tool-selection complexity	No single platform reported as universally optimal	Context-based selection aligned to objectives and competencies	[12]
Regulatory and standardisation uncertainty	Absence of uniform standards for algorithmic auditing	Standardised algorithmic-audit protocols	[13]

can be evaluated without legal exposure. Fifth, sector-specific implementation frameworks are needed, particularly for public-sector and non-profit organisations and SMEs, whose resource and staffing constraints differ from those of large regulated institutions. Finally, organisations should embed analytical outputs into formal governance so that identified anomalies inform upstream policy and controls.

C. Limitations and Future Research

The findings are qualified by several limitations. First, this is an integrative review of secondary sources and contributes no primary empirical data; its conclusions reflect the state of the reviewed literature rather than independently measured outcomes, and no causal effects can be established. Second, the evidence base is heterogeneous, combining peer-reviewed research with industry and professional reports; although sources were tagged by type, claims resting on grey literature are weaker than those supported by peer-reviewed studies. Third, the review is subject to publication and selection bias, and coverage is weighted towards English-language sources, which may under-represent practice in other regions and languages. Fourth, the database set, while broad, is not exhaustive and excludes some specialist venues. Fifth, the proposed prioritisation framework is conceptual and unvalidated: its variables, weights and threshold are specified for clarity and testability, not as empirically fitted parameters, and the reviewed techniques are reported individually rather than as an integrated system. Sixth, the rapid pace of change in analytics and generative AI means that some tool capabilities described here will date quickly.

These limitations define a research agenda. Future work should (i) validate the prioritisation framework empirically using real audit datasets, reporting precision, recall and false-positive rates against a baseline; (ii) test its governance properties, including explainability and fairness, particularly under the bias concerns noted above; (iii) conduct longitudinal, multi-sector field studies to assess how analytics maturity translates into audit outcomes; and (iv) investigate the organisational conditions—skills,

data governance and change management—that the synthesis identifies as decisive but that the reviewed literature does not evaluate jointly.

Bibliographic details were verified against public sources. Entries marked [†] could not be fully confirmed and require author confirmation before submission.

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